



History, Science, and Psychology

IN 1877 JAMES WARD AND JOHN VENN PETITIONED THE UNIVERSITY OF Cambridge in England to have experimental psychology introduced as an academic discipline. The University Senate refused to do so on the grounds that it would “insult religion by putting the soul on a pair of scales” (Hearnshaw, 1989, p. 125). In a 1907 paper published in *American Medicine*, Dr. Duncan Macdougall of Haverhill, Massachusetts, described his attempt to put the soul on a scale (Macdougall, 1907). He persuaded six dying patients to spend their last hours in a special bed that rested on a platform beam scale. By comparing the weight of the individual (plus bed) before and immediately after death, Macdougall estimated the weight of the human soul to be about “three-fourths of an ounce.” He repeated this experiment with 15 dying dogs, who manifested no weight loss upon expiration, confirming the popular belief that animals have no soul (Roach, 2003).

From the dawn of recorded civilization, humans have not only speculated about the nature and causes of mind and behavior, but have also employed their ingenuity to put these speculations to empirical test. In the seventh century BCE, the Egyptian King Psamtik I supposed that children with no opportunity to learn a language from other people would spontaneously develop the natural and universal language of humankind, which he presumed to be Egyptian (Hunt, 1994). He tested this hypothesis by having one of his subjects seclude a number of infants and observe which language they first spoke; he was disappointed to learn that they did not speak Egyptian. As the centuries progressed, critical thinkers continued to speculate about the nature and causes of mind and behavior and to subject their theories to empirical test. The process was accelerated by the scientific revolution in Europe in the 16th and 17th centuries and by the development of experimental physiology and evolutionary theory in the 19th century, which promoted the growth of the institutional science of psychology in the late 19th and 20th centuries. The story of this progression, development, and growth is the history of psychology.

WHY STUDY THE HISTORY OF PSYCHOLOGY?

In high school many years ago, I was an enthusiastic student of history and did fairly well in the subject. Once, when persuaded to serve on a student opinion panel to pass the days between final examinations and summer vacation, I was asked the question “Why study history?” At a loss for an answer, I responded “Because it’s there!”

In later years I came to recognize other answers. By studying the past we are better able to anticipate the future. Or, as the philosopher George Santayana put it, “Those who cannot remember the past are condemned to repeat it” (1905, p. 284). Unfortunately, knowledge of past errors is no guarantee against their repetition, as is attested by the continuing horrors of war. History also broadens our intellectual horizons, by introducing us to historically distant social and cultural forms of life, in much the same way as anthropology broadens our intellectual horizons by introducing us to alternative contemporary social and cultural forms of life. As in anthropology, however, there is always the danger of interpreting historically or culturally different forms of life in terms of our own cultural categories.

Still, both answers are legitimate reasons for studying the history of psychology. By learning about the overreaching ambitions of early American psychologists with respect to mental testing and eugenics, which supported restrictive immigration programs and compulsory sterilization, we hopefully insulate ourselves against similar ambitions. By learning about the theoretical visions of early psychologists such as René Descartes (1596–1650), Wilhelm Wundt (1832–1920), and John B. Watson (1878–1958), we come to appreciate the radically different ways in which past theorists conceived of mind and behavior.

In the case of the history of scientific disciplines such as psychology, there is another justification, which also answers a question naturally expressed by students: Why does the contemporary psychologist need to study the history of psychology? While it is possible to function as a psychologist without knowing the history of psychology, there are real dangers in doing so. Practitioners who neglect the history of their discipline fail to appreciate the historical contingency of the assumptions that have shaped their discipline, often via peculiar and accidental combinations of social circumstance and personality. This is especially true of the history of psychology, which in recent centuries has been significantly shaped by assumptions about the defining features of scientific thought.

Still, there was some point to my original response. The best reason for studying the history of psychology (like the history of anything) is because of its intrinsic interest. Anyone who finds psychological theory and experiment interesting, and most students who take courses in the history of psychology already do so, cannot fail to be engaged by the history of psychology.

Internal and External History

Contemporary **historiography**, the theory and methodology of history, recognizes a variety of approaches to the history of disciplines such as psychology. Traditional histories of psychology, such as Edwin G. Boring's (1886–1968) classic *A History of Experimental Psychology* (1929), have tended to be **internal histories**, largely devoted to the development of psychological theories and methods within the discipline. Such histories are generally written by “insiders,” that is, by psychologists themselves, and are thus sometimes characterized as “house histories” (Woodward, 1987). In contrast, more recent histories have tended to be **external histories**, which aim to account for the development of psychological science in terms of social, economic, political, and cultural conditions that promoted certain forms of psychological theory and practice and constrained others (Buss, 1975; Furumoto, 1989). Some of these histories have also been written by “outsiders,” that is, by professional historians rather than psychologists (e.g. Smith, 1997), although this remains relatively rare.

Of course, few histories of psychology adopt an exclusively internal or external approach, and the appropriate form of historical analysis ought to be determined by the historian's judgment about whether internal or external factors played a more influential role during any significant period (cf. Boakes, 1984, pp. xiii–xiv). For example, the different internal intellectual traditions of Great Britain and Germany probably best explain the differences between British associationist psychology and the German holistic psychology of Wilhelm Wundt (1832–1920) and the Gestalt psychologists. In contrast, external factors such as the pragmatic and utilitarian orientation of turn-of-the-century America clearly play a major role in accounting for the development of functionalist and behaviorist psychology in America in the early decades of the 20th century. Yet this can scarcely be the whole story, since institutional psychology also became increasingly applied in Germany and France at around the same time.

Zeitgeist and Great Man History

Histories of psychology also differ in how much influence they attribute to major psychologists, or **great men**, as opposed to the **zeitgeist**, or “spirit of the times” (Boring, 1929). Again, how much attention ought to be paid to either factor ought to be determined by the historian's judgment about the respective influence of these factors during any historical period. While Wundt deserves credit for founding the first experimental laboratory at the University of Leipzig in 1879, it may be argued that psychology would have developed in Germany in much the same way that it did if Wundt had never lived. On the other hand, although behaviorism no doubt would have developed in America even if John

B. Watson had never lived, it likely would not have taken the specific form that it did in the 1920s.

Sometimes a major historical development is a product of both a significant individual and the spirit of the times, of someone being the right person in the right place at the right time. Ivan Pavlov (1849–1936) is famous for his “discovery” of what is now known as classical conditioning. He demonstrated that the salivatory reflex of dogs and other animals could be conditioned to the presentation of a neutral stimulus when it is regularly paired with food. Yet this form of learning was identified centuries earlier. For example, it was described by the Edinburgh physician Robert Whytt (1714–1766), who cited conditioned salivation (to the smell of a lemon) as an illustration. Edwin B. Twitmyer (1873–1943), an early pioneer of speech pathology, discovered that the patellar (knee-jerk) reflex could be classically conditioned and made it the subject of his doctoral dissertation at the University of Pennsylvania. When he completed his thesis, “A Study of the Knee Jerk,” in 1902, he arranged to have it published privately, but it attracted little attention. Twitmyer recognized the significance of this form of conditioned learning and delivered a paper on his research at the 1904 meeting of the American Psychological Association, but it fell on deaf ears.

It was only with Pavlov’s investigations that this form of learning was adopted as an explanatory paradigm by behaviorist psychologists. Pavlov had the scientific prestige, having won the Nobel Prize in physiology for his work on digestion. His investigations were based upon rigorously controlled experiments conducted by a team of researchers at a scientific institute, at a time when rigorous experimentation was treated as the distinctive mark of genuinely scientific psychology. Pavlov’s work became known in translation to American psychologists at precisely the time when they were developing explanations of animal and human behavior in terms of correlations between observable stimuli and responses (Logan, 2002).

Sometimes the influence of certain psychologists is a product of fortuitous circumstances. Thus Watson was fortunate to attain the chairmanship of the Johns Hopkins department of psychology and the editorship of *Psychological Review* as a result of James Mark Baldwin’s (1861–1934) being forced to resign these positions after being arrested in a brothel. Clark L. Hull’s (1884–1952) neobehaviorist theories of learning may have been superior to those of Edward C. Tolman (1886–1959), but the success of Hull’s research program was at least partly due to the fact that it received generous funding from the Laura Spelman Rockefeller Memorial Fund during the Depression, whereas Tolman’s did not.

Presentist and Contextualist History

Historians also distinguish between what has been called **presentist history** of psychology, sometimes also known as “Whig” history, in which the history of psychology is represented as approaching and approximating (idealized) contemporary

theory and practice, and **contextualist history**, sometimes also known as “historicism,” in which each historical episode or epoch is explicated neutrally in its own terms (Stocking, 1965). Presentist approaches have long been popular and generally represent the history of psychology as a long evolution from primitive theories about immaterial souls or spirits to the modern scientific endeavor. Yet although it is certainly true that many early theorists believed in immaterial souls or spirits, it does a great injustice to pioneers such as Hippocrates (c. 460–377 BCE) and Descartes to represent early psychologists as primitive thinkers.

The Greek physician Hippocrates rejected traditional accounts of epilepsy in terms of spirit possession and advanced his own account in terms of brain damage and dysfunction. Although Descartes did maintain that the mind is an immaterial substance, he also proposed the first systematic reflex theory of animal and (some) human behavior. Medieval Christians did not burn hundreds of thousands of psychotics and schizophrenics whom they ignorantly misdiagnosed as witches. Rather, 17th- and 18th-century hysterics in Europe and America did that, *after* the scientific revolution in Europe that began in the 16th century. Indeed, it may be reasonably argued that the persecution of witches in Europe and America was itself largely a product of the scientific revolution of the 16th century (Cohen, 1975; Kirsch, 1978).

Although the general movement from a “primitive” to an empirically based psychology marked an intellectual advance, the development of scientific psychology did not proceed in as smooth or linear a fashion as is normally supposed. Indeed, one may reasonably maintain that at certain critical periods, including the 20th century, psychological science regressed. For example, Aristotle is conceptually closer to contemporary cognitive psychologists than many of the late-19th- and early-20th-century pioneers of psychological science.

On the other hand, there are serious problems associated with contextual approaches that profess to adopt a completely neutral attitude to the history of psychology. It is certainly appropriate, for example, to try to explain why behaviorism appealed to many American psychologists in the 1920s: to try to explain why, given their intellectual and social institutional background, it was *reasonable* for many psychologists to adopt behaviorism in the 1920s. However, it is doubtful that one can determine the significance of this important episode in the history of psychology without some working conception of the nature and potential of psychological science and thus of whether the behaviorist period represented an advance or regression in the general development of psychological theory and practice.

Conceptual History of Psychology

While historians of psychology have vexed over these historiographic matters, they have tended to neglect another project. This is the identification of significant conceptual continuities and discontinuities in the history of psychological theory and practice, such as the conceptual continuity between the approaches of

Aristotle and contemporary cognitive psychology and the conceptual discontinuity between “liberalized” neobehaviorist theories and those of contemporary cognitive psychology. Without some grasp of these continuities and discontinuities, any explanatory history of psychology is theoretically blind. In the **conceptual history** of psychology that follows, I focus on these continuities and discontinuities, offering explanations of thematic developments based upon contemporary scholarship.

The history of psychology is still in its infancy as an academic discipline. Although the first histories of psychology were written in the early decades of the 20th century (Baldwin, 1913; Brett, 1912–1921), the history of psychology became established as a subdiscipline of psychology only in the 1960s, with the founding of the *Journal of the History of the Behavioral Sciences* in 1965 and the establishment of the Division of the History of Psychology of the American Psychological Association that same year. Cheiron: The International Society for the History of the Behavioral and Social Sciences was formed in 1969; the NSF summer institute that led to its formation was held at the University of New Hampshire in 1968, where the first PhD program in the history of psychology was instituted. Consequently, the explanations in this work should be recognized as partial and tentative and



Participants at the NSF summer institute at the University of New Hampshire in 1968, which led to the formation of Cheiron: The International Society for the History of the Behavioral and Social Sciences in 1969.

relative to the level of analysis. Deeper levels of analysis may reveal richer conceptual strands, and readers are encouraged to pursue them.

While the early history of psychology ranges over the Mediterranean, the Middle East, and Europe, and 19th-century history focuses upon developments in Britain, France, and Germany, the 20th-century history of psychology is very much the history of American psychology. Although institutional scientific psychology originated in Germany at the end of the 19th century, by the beginning of the 20th century American psychology came to dominate other national psychologies in terms of the number of psychologists, institutions offering degrees, books, journals, and student populations. It maintained its dominance throughout the 20th century (Brandt, 1970; Koch, 1992; Rosenzweig, 1984), especially after the Second World War, when it effectively “colonized” the national psychologies of many European states (van Strein, 1997) and Japan.

SCIENCE AND PSYCHOLOGY

One of the distinctive features of early scientific psychology and later forms of academic psychology is the degree to which they were shaped by prevalent conceptions about the nature of science. Psychology, perhaps more than any other discipline, self-consciously modeled itself upon successful sciences such as physics, chemistry, and biology. In consequence, many contemporary psychologists embrace a number of principles that are of questionable relevance to psychological science.

To illustrate this important point, it is useful to distinguish between those principles that are generally agreed to be essential features of empirical science (as opposed to formal sciences such as logic and pure mathematics) and those principles whose relevance is an open question.

Objectivity

It is generally acknowledged that a minimal condition of an intellectual discipline constituting a science is that the propositions it offers are **objective**. Propositions are objective if their truth or falsity is determined by independent facts. Thus, the propositions that *bodies of different weight fall with equal acceleration* and that *electrons have a negative electric charge* are objective because they are true if and only if bodies of different weight do fall with equal acceleration and electrons do have a negative electric charge (and false otherwise). Analogously, the propositions that *the patellar reflex can be classically conditioned* and that *humans employ prototypes in category formation* are objective because they are true if and only if the patellar reflex can be classically conditioned and humans do employ prototypes in category formation (and false otherwise).

The objectivity of scientific propositions should be distinguished from the objectivity of the judgments of scientists about the best theories in any domain (the best theories of molecular bonding, neural transmission, or human aggression, for example). Such judgments are objective if they are *unbiased*, and **subjective** if they are biased by individual or collective preferences or by social, political, or religious interests in the advocacy of certain theories (for example, that the earth is the center of the universe, that evolution is progressive, or that there are racial and gender differences in intelligence). The objectivity of propositions is also not equivalent to **materialism**: the view that ultimate reality is material. Although many propositions are rendered true or false by independent facts about the existence (or nonexistence) and properties of material bodies, others may be rendered true or false by independent facts about abstract objects such as numbers or by independent facts about the existence (or nonexistence) of immaterial or spiritual entities such as immortal souls or a benevolent God.

Causal Explanation

Of course, the requirement of propositional objectivity does not distinguish the propositions of scientific disciplines from those of everyday life or religion. The propositions that *cats like milk* and that *God is good and all powerful* are likewise objective because they are also true if and only if there are cats that like milk and there is a God that is good and all powerful (otherwise they are false). Another essential requirement of a scientific discipline, and one that goes some way to distinguishing scientific disciplines from other forms of speculation, is *causal explanation*: the propositions of scientific disciplines advance causal explanations of how certain events, regularities, or structures are generated or produced. Thus biologists explain patterns of embryonic development in terms of genetic programming, and psychologists explain systematic errors in probabilistic reasoning in terms of cognitive heuristics.

Causal explanations of classes of events, regularities, or structures cite factors that are held to be conditions for them: Their existence is held to be *conditional* upon the prior (or simultaneous) existence of such factors. To explain rusting in terms of oxidation is to claim that the presence of oxygen is a condition for rusting; to explain learning in terms of reinforcement is to claim that reinforcement is a condition for learning. All causal explanations cite conditions that are held to be sufficient for the generation of an effect, given other enabling conditions. Such conditions are sometimes also held to be necessary for the generation of an effect, but not always. For example, a source of ignition is often held to be both necessary and sufficient for combustion, given other conditions such as the presence of oxygen. However, the presence of a violent stimulus is not held to be necessary for aggressive behavior, even if it is held to be sometimes sufficient, given other

enabling conditions (Berkowitz & Le Page, 1968), since there are other recognized causes of aggressive behavior such as frustration and anger.

Causal explanations are often couched in terms of functional relations between variables, when one variable is held to increase or decrease with another: Thus the increased volume of a gas (at constant pressure) is held to be functionally explained in terms of increased temperature, and increased levels of “destructive obedience” are held to be functionally explained in terms of the increased proximity of commanding authorities (Milgram, 1974).

Empirical Evaluation

Of course, everyday folk also offer causal explanations of events, regularities, and structures, so an appeal to causal explanation is insufficient to distinguish scientific physics and psychology from so-called folk-physics and folk-psychology. What does distinguish most folk descriptions and explanations from scientific ones is that the latter are subject to empirical evaluation. Scientific descriptions and explanations are tested either directly by observation or, in the case of theoretical descriptions and explanations about unobservables such as electrons or repressed thoughts, indirectly via their observational implications. This condition goes a long way to account for the fact that scientific disciplines are also generally held to be objective in the sense that the judgments of scientists are *unbiased*. Systematic methods of empirical evaluation, including experimentation, are held to enable scientists to adjudicate between alternative causal explanations independently of personal, social, political, or religious biases. Thus, properly scientific judgments are held to be adjudicated (ideally) by empirical data alone.¹

Karl Popper (1963) has claimed that the testability, or **falsifiability**, of scientific theories is what distinguishes genuine sciences such as physics and biology from **pseudosciences** such as astrology, psychoanalysis, and Marxism. According to Popper, genuine scientific theories do not merely accommodate known empirical data, but make risky predictions which, if falsified by consequent observations, would lead to their rejection; an example is Einstein’s risky prediction that light rays traveling to Earth from distant stars would be deflected by the gravitational force of the sun (a prediction corroborated by Eddington in 1917). In contrast, pseudosciences either do not generate risky predictions (in the case of astrology and psychoanalysis) or accommodate failed predictions by ad hoc modifications of the theory designed to protect it from falsification (in the case of Marxist theory).

The practice of phrenology in the early 19th century was frequently pseudoscientific. Franz Joseph Gall (1758–1828) and his followers maintained that innate

¹In conjunction with other theoretical desiderata such as simplicity, fertility, and the like.



Skull with phrenological markings.

psychological faculties are manifested as protrusions or indentations of the skull, caused by the over- or underdevelopment of the region of the brain associated with the faculty. Phrenologists were quick to seize on evidence that supported their theoretical localization of psychological faculties to regions of the brain, but explained away negative evidence in a variety of ad hoc ways. For example, brain protrusions not associated with superior development of the associated faculty were explained away in terms of brain damage. Original estimates of the development of a psychological faculty were sometimes revised to accord with those predicted by the degree of protrusion or indentation of the skull. When phrenologists discovered that Descartes' skull was indented in the area of the brain where the rational faculty was supposed to be located, phrenologists concluded that Descartes could not have been as great a thinker as was commonly supposed (Young, 1990, p. 43).

While these three conditions seem clearly necessary for any scientific discipline, it may be doubted whether they are sufficient. It may be argued that scientific explanations should also be quantitative: that they should describe mathematical relations between variables, such as Boyle's law (at constant temperature the volume of a given mass of gas is inversely proportional to its pressure) or Fechner's law (the intensity of a sensation is a logarithmic function of the intensity of the physical stimulus). Yet although quantification is normally a virtue, it is doubtfully necessary, since empirically well-supported qualitative explanations,

such as Charles Darwin's (1809–1882) account of the evolution of species through natural selection (Darwin, 1859) and Jean Piaget's (1896–1980) account of cognitive development in children (Piaget, 1926) seem valuable contributions to scientific knowledge. It may be argued that a certain amount of systematicity is also required: that scientific explanations should fit within some coherent general theory, such as the explanations of the properties of the elements that make up the periodic table in terms of differences in their composition and structure. However, it is difficult to specify what this amounts to and seems to dogmatically presuppose some predetermined degree of system in nature (including human nature).

Moreover, even if such conditions were considered sufficient for a discipline to count as scientific, many questions would remain unanswered. One such question concerns the nature of causality and causal explanation. Is causality more than conditionality, and do causal explanations do anything more than cite empirical conditions that enable scientists to predict empirical outcomes, such as combustion and conformity? How does causation relate to correlation? It is generally agreed that causality is not equivalent to correlation, even though the identification of causality is based upon the observed correlation of conditions and effects. Two factors may be highly correlated, but not causally related, because they may be joint effects of independent conditions (the propensity to watch violent television and engage in aggressive behavior may be joint effects of childhood abuse) or cyclical processes that happen to be sequentially related (such as the correlation between the population of mules and PhD students in southern California, which rise and fall together). Conversely, some causal conditions may be rarely correlated with their effects, because of interference (lead screening ensures that humans are rarely affected by plutonium sickness, and parents and the police may discourage aggression in children inclined to it by exposure to violence on television or the street).

The distinction between experimental studies, designed to identify causal conditions in artificially isolated and controlled “closed” systems, and correlational studies, designed to identify the degree of correlation between factors in naturally occurring “open” systems (Bhaskar, 1975), was popularized in psychology through Robert Session Woodworth's (1869–1962) *Experimental Psychology* (1938), and institutionalized in “The Two Disciplines of Scientific Psychology,” Lee J. Cronbach's presidential address to the American Psychological Association in 1957.

Another question concerns the status of theoretical claims about unobservable entities such as electrons or motives. Should theoretical claims about such entities be treated as potential descriptions of possibly real entities or as merely useful (or useless) fictions? **Realism** is the view that scientific theories about entities that are not observable—or not directly observable (in the case of electrons) or intersubjectively observable (in the case of motives)—are potentially true descriptions of them. **Instrumentalism** is the view that scientific theories “about” unobservable entities such as electrons or motives are not potentially true descriptions of

them, but merely linguistic devices that facilitate the prediction of the behavior of observable entities, such as electric circuits or human behavior. The realist view holds that there are objective conditions for the truth of theoretical claims over and above the truth of the empirical predictions that can be derived from them. The instrumentalist view holds that there are no such conditions: The truth of the empirical predictions that can be derived from theoretical claims is sufficient for the “truth” of these theoretical claims. Realism has been the favored position among natural scientists, but instrumentalism was a popular position among 20th-century neobehaviorist psychologists (e.g., Kendler, 1952).

Some of these questions are considered in later chapters of this work. However, it is not important for our present purposes to provide a complete definition of science or final answers to these questions. What is important is to distinguish essential features of science from a set of principles that are frequently associated with science but cannot be considered essential to it. Many psychologists adopted these principles, which embody assumptions about the subject matter and scope of explanations in science, because they were associated with early exemplars of successful physical science, even though it is an open question whether they are appropriate for psychological science. One of the aims of this work is to document how psychologists came to adopt these principles.

Atomism

One of the principles associated with science is **atomism**, which holds that the entities that form the subject matter of scientific disciplines can be individuated and exist independently of other entities to which they may be related. That is, they can be theoretically described without making reference to other entities and can exist in the absence of (or in isolation from) other entities. This principle holds for elements such as carbon, which can be theoretically described in terms of its composition, structure, and properties without citing any other elements or their properties. Carbon could in principle exist even if no other element existed, and samples of carbon can be isolated from other elements to which they may be related (causally or spatially).

However, this principle does not hold for entities such as quarks (the constituents of protons, neutrons, and electrons) or parts of electromagnetic fields, which appear to be **relational** in nature: They can be individuated and exist only in relation to other entities. Individual quarks or parts of electromagnetic fields can be theoretically described only by reference to other quarks or parts of electromagnetic fields, and individual quarks or parts of electromagnetic fields cannot be isolated from other quarks or parts of electromagnetic fields. For this reason sciences such as physics have abandoned the principle of atomism.

Many psychologists have assumed that psychological states and behavior are atomistic in nature, the notable exception being the Gestalt psychologists.

They have assumed that psychological states and behavior can be theoretically described and experimentally isolated independently of their relation to other psychological states and behavior. However, it may be reasonably doubted whether this principle holds for certain psychological states and behavior. Cognitive states such as beliefs seem to be relational in nature, because they presuppose a network of other beliefs. It may be doubted, for example, that a person could be ascribed only a *single* belief, such as the belief that the Empire State Building is in New York City. The true ascription of such a belief would appear to presuppose that the person has other beliefs about New York City (such as where it is), a grasp of the semantics of the linguistic contents of the belief (what the terms *building* and *in* mean, for example), and so forth. Analogously, certain forms of social behavior, such as serving on a jury or engaging in altruistic or aggressive behavior, seem to be relational in nature: They appear to presuppose an institutional context and relationship to other persons. However, it ought to be stressed that the question of whether psychological states and behavior are atomistic or relational in nature (or the degree to which they are atomistic or relational) is an open question. The point is only that there is nothing unscientific about supposing that some psychological states and behavior are *not* atomistic in nature.

Universality of Causal Explanation

Another principle associated with science is the **universality of causal explanation**, sometimes known as the *singularity* of causality. According to this principle, the same causal explanation applies to each and every instance of a class of events, regularities, or structures. This seems to be true of rusting, superconductivity, and biological death, which appear to have only one kind of cause. However, it is not obviously true of physical motions, which may be caused by either gravitational or electromagnetic (or strong or weak nuclear) forces, or of some cancers, which may be caused by either genetic or environmental factors.

Nevertheless, from the time of Isaac Newton (1642–1727) to the present day, psychologists have regularly insisted that universality is the measure of the scientific adequacy of psychological explanation (Kimble, 1995; Shepard, 1987, 1995): They have assumed that there is one and only one causal explanation of aggression, depression, or learning, for example. However, it may be reasonably supposed that some psychological states and behaviors have more than one cause. It does not appear unscientific or absurd to suppose, for example, that some aggressive behaviors are products of motives of revenge, whereas others are caused by the presence of “violent stimuli” such as weapons (Berkowitz & Le Page, 1968) and others by overexcitation of the lateral hypothalamus (brought on by drugs or diet). It does not appear unscientific to suppose that some forms of depression

are the product of genetic predisposition and others a function of environmental pressures. Again, it should be stressed that it is an open question whether aggression or depression do have more than one cause. The point is that there is nothing unscientific about supposing that they do.

Ontological Invariance

A closely related principle is **ontological invariance** in space and time. According to this principle, the kinds of entities that constitute the subject matter of scientific disciplines can be re-identified in all regions of space and time. This principle appears to hold for fundamental physical particles and forces, which we believe to have been around for all time (or at least since the big bang) and to be found in all regions of space, and possibly also for many chemical elements and compounds. However, it does not appear to hold for organic life forms, some of which are later evolutionary developments and some of which are not found in many regions of space (for example, on planets too hot or too cold to sustain them). Thus, while fundamental branches of physics and chemistry embrace this principle, sciences such as biology do not, since species and viruses transform themselves (and become extinct) in historical time and are not to be found in all regions of the earth (far less the universe).

Once again, the point is not to prejudge open questions, but to note that there is nothing unscientific about supposing that certain entities are not invariant in space and time. Consequently, there is nothing unscientific about supposing that certain psychological states and behaviors are not invariant in cultural space and historical time. For example, it appears that the behavioral practice of *couvade*, in which husbands empathetically simulate the birth pangs of their wives, may be unique to a small number of Amazonian tribes. The emotion of *amae*, a kind of “fawning” dependency, may be distinctively Japanese (Doi, 1973), and *fago*, a complex emotion involving elements of death, going on a journey, and being in the presence of an admirable person, may be unique to the Ifaluk (Lutz, 1982). The pathological emotion of *accidie*, a debilitating form of disgusted boredom, may have been restricted to medieval times (Altschule, 1965).

Although natural scientists have been prepared to abandon the principle of ontological invariance, psychologists have been reluctant to do so. Indeed, many contemporary psychologists oppose the notion that psychological explanation may vary cross-culturally and transhistorically because the psychologies of different cultural and historical communities may be distinct. The suggestion that there might be “indigenous psychologies” localized to specific cultural or historical communities (Heelas & Lock, 1981; Moghaddam, 1987) has met with a vigorous critical response from psychologists (Kimble, 1989; Staats, 1983; Spence, 1987), many of whom have insisted that any form of psychology that implies the cultural or historical restriction of psychological explanation is unscientific.



Testing for cross-cultural differences in visual acuity (The Torres Strait Expedition, 1898).

Of course, it is often very difficult to determine whether a certain emotion such as *accidie* or a disorder such as schizophrenia varies cross-culturally or transhistorically, since *accidie* may be present in contemporary cultures even though they do not have a word for it (Findley-Jones, 1986), and the identification of schizophrenia in medieval times is hampered by the limited availability of clinical descriptions from that period (Heinrichs, 2003). However, acknowledging that it is difficult to determine cross-cultural and transhistorical variance in psychology and behavior does not mean that it is unscientific to suppose it exists.

Explanatory Reduction

Another influential principle is **explanatory reduction**, according to which the best explanation of a complex entity, property, or process is given by an analysis of its material components. This principle has served some physical sciences very well. The causal properties of the elements of the periodic table are best explained in terms of their electronic components and chemical bonds, and the thermodynamic properties of gases are best explained in terms of statistical mechanics (by treating gases as collections of molecules in random motion). Yet not all physical scientific explanations proceed in this fashion. Sometimes the best explanation

operates by specifying relations on the same level as the entities or processes explained. Thus contemporary physics does not treat the mass of a physical body as a function of the autonomous masses of its components, but as a function of its relation to other physical bodies. Evolutionary processes in contemporary biology are partially explained in terms of genetics and partially explained in terms of the environments in which organisms are situated.

The same is true of psychological science. It might turn out that neurophysiology and biology will ultimately furnish the best explanations of mind and behavior, as generations of theorists have hoped and anticipated. Yet this might not turn out to be the case. The best explanations might turn out to be those that develop theories of the relational integration of our cognitive architecture and behavior. The development of Sigmund Freud's (1856–1939) theory of neurosis is an interesting example. Freud was originally trained as a physiologist and developed his theory of neurosis in terms of repressed memories after he realized he could not provide a reductive physiological explanation of *conversion hysterias*: instances of physical paralysis in which there is no discernible physiological abnormality. Freud explained these cases of paralysis by treating them as manifestations of anxiety symbolically related to past traumatic episodes. As Freud put it in his paper, "The Unconscious":

Research has given irrefutable proof that mental activity is bound up with the function of the brain as it is with no other organ. . . . But every attempt to go from there to discover a localization of mental processes, every endeavor to think of ideas as stored up in nerve-cells and of excitations as traveling along nerve-fibers, has miscarried completely. . . . *Our psychical topography has for the present nothing to do with anatomy*; it has reference not to anatomical localities, but *to regions in the mental apparatus*, wherever they may be situated in the body.

—(1915/1957, pp. 174–175, my emphasis)

Of course one might question the adequacy of Freud's account of conversion hysterias, and Freud himself originally tried to develop a reductive physiological theory (which he later abandoned) in "Project for a Scientific Psychology" (1895/1950). But once again, the point is only that there is nothing unscientific about the supposition that the best explanation of psychological states and behavior might not be reductive.

Determinism

Another principle much favored by psychologists is **determinism**. According to this principle, for every event there is a set of prior conditions whose combination is sufficient to generate that event, such that no other outcome is possible. For example, if a billiard ball is struck by another and moves off with a certain velocity,

it is presumed that it does so because the force of the ball colliding with it is sufficient to propel it and, given the force of the colliding ball, no other outcome is possible. Although physical scientists embraced this principle for many centuries, contemporary physicists have abandoned it. According to contemporary quantum mechanical theory, radioactive decay is not uniquely determined by prior conditions: These conditions ensure that there is a certain (fairly high) probability that a beta particle will be emitted, but do not ensure that a beta particle will be emitted. Analogously, it does not appear to be unscientific to suppose that the conditions responsible for human aggression or depression merely incline or promote (or render probable) instances of aggression or depression without determining them.

Psychologists are strangely reluctant to embrace such a possibility, insisting that determinism is a presumption of science. Yet for most practical purposes it does not matter whether psychologists embrace this principle, since psychologists would advance and test explanations in much the same way whether they thought conditions determine or simply promote psychological or behavioral outcomes. They would proceed in the same way whether they thought “violent stimuli” determine or merely incline people to be aggressive, for example. They would still predict that people would tend to become aggressive in the presence of violent stimuli and that there would be statistically significant differences between the behavior of subjects who are exposed to “violent stimuli” and those who are not.

Two other features are often treated as essential to science. The first is commitment to experimentation as the mark of a genuinely scientific discipline, and the second is commitment to empiricism.

Experimentation

Experimental sciences are those in which scientists are able to create situations in which causal conditions can be isolated and causal explanations evaluated. For example, at Camp Lazear in Havana in 1900, Walter Reed and James Carroll determined that the bite of the tiger mosquito is the cause of yellow fever by experimentally isolating human volunteers and exposing them to “noxious vapors” from swamps, contact with fellow sufferers (strictly speaking, their soiled clothing), and tiger mosquitoes—the three then-prevalent hypotheses about the cause of yellow fever. Since after 30 days in isolation, only subjects exposed to tiger mosquitoes contracted yellow fever, the researchers concluded that tiger mosquito bites are the cause of yellow fever and that contact with “noxious vapors” and contagion are not.

The logic of experimentation is directly related to the conception of causality as conditional. Given this conception, scientists attempt to identify conditions that are regularly present when a certain effect is present and regularly absent when a certain effect is absent or to identify variables that regularly increase or decrease when other

variables increase or decrease. John Stuart Mill's (1806–1873) methods of agreement, difference, and concomitant variation, often known as “Mill's methods” (Mill, 1843), describe situations in which such conditions can be identified.

According to Mill's **method of agreement**, if instances of an effect have only one condition in common, then that condition is the cause of the effect. According to **Mill's method of difference**, if an instance in which an effect occurs and an instance in which it does not occur differ with respect to only one condition, then that condition is the cause, or an essential part of the cause, of the effect. According to the **method of concomitant variation**, if one condition increases or decreases while an effect increases or decreases, then that condition is the cause of the effect.

Since correlation is not equivalent to causality, no amount of positive instances of correlation between conditions and effects—between, for example, a form of psychological treatment and the elimination or attenuation of neurotic symptoms—can establish a causal connection. However, one negative instance of the presence of a condition in the absence of an effect—a case, for example, of excessive masturbation not followed by blindness—can demonstrate the absence of a causal connection. The method of eliminating competing causal hypotheses until only one viable hypothesis remains is known as **eliminative induction**, a method Francis Bacon (1561–1626) promoted during the scientific revolution in Europe. According to this method, it is not sufficient to observe that yellow fever is commonly preceded by exposure to tiger mosquito bites to establish that tiger mosquito bites are the cause of yellow fever. One has first to eliminate alternative causal hypotheses in terms of exposure to noxious vapors and contagion, by demonstrating the absence of yellow fever in the presence of noxious vapors and contact with other victims of yellow fever. Analogously, it is not enough to observe elimination or attenuation of neurotic symptoms commonly preceded by a form of psychological therapy to establish the causal efficacy of that form of psychological therapy. One must first eliminate alternative causal hypotheses in terms of spontaneous remission (most neurotics get better anyway) or a placebo effect (engendered by client or therapist expectations of improvement).

Of course such characterizations are idealized. Since causal conditions can be counteracted by interference conditions (protective paint can prevent rusting, even if oxygen and water are present, and parents and the police can suppress aggressive behavior, even in the presence of violent stimuli), causal investigations presuppose that potential conditions of interference are absent in naturally occurring situations or have been eliminated or controlled for in experimental situations. Thus, experimental observations are usually preferred to naturalistic observations when they can be obtained, since experimental isolation and control can help to eliminate or attenuate potential interference conditions.

The ability to create experimental situations is a great convenience for sciences that are enabled to do so, but experimentation is not an essential feature of science, since causal explanations can be evaluated via forms of observation that

do not involve experimental manipulation and control. Many highly successful sciences such as astronomy and geology are not experimental sciences, and one may contrast the relatively few repetitions of controlled experiments commonly employed in physics and chemistry with the mass of observations patiently gathered in astronomy and ethology. However, scientific psychology was instituted as an academic discipline in Germany in the 1880s in large part through its commitment to experimentation, and generations of 20th-century psychologists have seen commitment to experimentation as the *sine qua non* of scientific psychology; this has been especially true of social psychologists (Greenwood, 1994). Robert S. Woodworth, who popularized the distinction between experimental and correlational studies in *Experimental Psychology* (1938), also promoted the notion that experimentation is the best, if not the only, reliable means of evaluating causal explanations (Winston & Blais, 1996).

Many psychological states and behaviors can be objects of experimental analysis, yet it is not obvious that they all can. Although psychologists clearly have the ability to manipulate and control human psychology and behavior, it is not clear that they can experimentally isolate all aspects of it. In the case of social-psychological states and behavior, for example, it may not be possible to re-create social attitudes or jury behavior in isolation from their everyday social contexts. As Chapanis (1967, p. 558) put it, with respect to many social-psychological states and behavior, “the very act of bringing a variable into the laboratory usually changes its nature.”

The familiar definition of experiments in terms of the active manipulation of “independent variables” is a 20th-century psychological construction, common to textbooks of psychology but rarely found in textbooks of physics and biology (Winston & Blais, 1996). The source of this definition is Edwin G. Boring’s *The Physical Dimensions of Consciousness* (1933, pp. 8–9), although it was popularized in psychology in Woodworth’s *Experimental Psychology* (1938), known as the “Columbia Bible” (Winston, 1990).

Empiricism

Science is often held to be founded on empiricism. In one clear sense it is. The principle of **methodological empiricism** requires that all scientific descriptions and explanations be subject to observational evaluation (the term *empiricism* comes from the Greek *empeirikos*, which means “experience”) and is just the third condition earlier identified as essential to science. In this respect, all scientists are empiricists. However, this principle should be carefully distinguished from **dogmatic empiricism**, the highly contentious view that scientific theory and causal explanation are restricted to the description of the correlation of observables.

It should also be carefully distinguished from the doctrine of **meaning empiricism**, according to which linguistic terms derive their meaning through

association with observables (that can be seen, touched, or otherwise experienced through the senses). The most contentious version of this doctrine, popular in psychology from the 1930s onward, is the doctrine that theoretical propositions “about” unobservable entities such as protons or repressed memories must be operationally defined in terms of observables. Despite its popularity among psychologists, the **operational definition** of the meaning of theoretical propositions is virtually unknown outside of the social sciences and thus cannot be reasonably maintained as an essential feature of science. Later chapters of this work chart the historical adoption of this doctrine by psychologists.

This doctrine became popular among psychologists in part because in its original form it was linked to a particular theory about the origin of concepts or ideas. According to the theory of **psychological empiricism**, all concepts or ideas are derived from experience. In this view, only someone who has experienced the color red can form the concept or idea of “red.”

Scientific Method

Psychological scientists regularly appeal to their employment of the scientific method in justifying the scientific status of their discipline, conceived either as a method of deriving theories from observations or as a method of postulating theories and testing them via their deductive or predictive implications. The former is usually characterized as the **inductive method**, the latter as the **hypothetico-deductive method**. Galileo Galilei’s (1564–1642) derivation of his law of falling bodies from his measurement of the velocities of balls rolling down an inclined plane and Jean Piaget’s descriptions of the stages of cognitive development based upon his observations of the development of his own children (Piaget, 1926) exemplify the inductive method. Johannes Kepler’s (1571–1630) postulation of elliptical orbits to explain planetary motions and Leon Festinger’s (1919–1989) prediction that subjects committed to certain beliefs will continue to maintain these beliefs in the face of contradictory evidence as a consequence of “cognitive dissonance” (Festinger, Riecken, & Schachter, 1956) exemplify the hypothetico-deductive method.

Francis Bacon is usually held to be the principal advocate of the inductive method, conceived of as a systematic means of ascending from observations to increasingly more general theories. Although some natural scientists and psychologists do seem to have developed their theories in this fashion, as in the case of Galileo and Piaget, later methodologists came to hold that such a method is not necessary and maintained that many successful theories are based upon hunches, lucky guesses, or prior speculation.

For example, Otto Loewi (1873–1961) had a peculiar dream. He imagined a tank of water in which two frog hearts were suspended. He dreamed that he

stimulated the vagus nerve of one heart, causing it to beat, and lo and behold, the second also began to beat! Perplexed but intrigued, he set about reproducing the conditions of his dream. He suspended two frog hearts in separate tanks of fluid connected by tubing, and found that stimulation of the vagus nerve of one produced heartbeat in both. He recognized immediately that some chemical transmitted from one heart to the other must have produced the stimulation of the second heart, and the theory of neurotransmitters was born.

It is doubtful that Galileo developed his law of falling bodies from observations of balls rolling down an inclined plane. The mathematical formulation of Galileo's law of falling bodies, the so-called mean-speed theorem, was stated by mathematicians associated with Merton College, Oxford, in the 14th century. Galileo admitted as much, claiming that he did his experiments "in order to be assured that the acceleration of heavy bodies falling naturally does follow the ratio expounded above" (1638/1974, p. 169), where the "ratio expounded above" is a proof of the Merton mean-speed theorem (Harré, 1981). Yet this does nothing to belittle Galileo's achievement, which was to empirically evaluate and consequently establish this formula.

Later scientific methodologists came to reject the Baconian notion of a "logic of discovery." John Herschel (1792–1871) distinguished between the "context of discovery" and the "context of justification" of scientific theories (Herschel, 1830). He noted that although many theories are formulated as a result of some form of inductive assent, the source of a theory is irrelevant to its scientific acceptability. Theories are accepted on the basis of their conformity with observation and experiment, and a lucky theoretical guess that is superior to an inductively derived theory in terms of its predictive success is always preferred. Herschel's contemporary, William Whewell (1794–1866), also maintained that many of the most significant advances in science, such as Kepler's postulation of elliptical orbits, were a product of "felicitous and explicable strokes of inventive talent" (1858, p. 64).

These critical responses evolved into the 20th-century position known as hypothetico-deductivism. According to this position, the source of a scientific theory is irrelevant to its empirical adequacy, which is a function of its confirmed empirical implications:

There are . . . no generally applicable "rules of induction," by which hypotheses or theories can be mechanically derived or inferred from empirical data. The transition from data to theory requires creative imagination. Scientific theories are not *derived* from observed facts, but *invented* to account for them. . . . Scientific objectivity is safeguarded by the principle that while hypotheses and theories may be freely invented and *proposed* in science, they can be accepted into the body of scientific knowledge only if they pass critical scrutiny, which includes in particular the checking of suitable test implications by careful observation and experiment.

—(Hempel, 1966, pp. 15–16)

According to this view, science develops through a process of hypothesis postulation and empirical testing or a series of “conjectures and refutations” (Popper, 1963).

Since the time of Herschel and Whewell, methodologists have also stressed the significance of **novel predictions** in establishing scientific theories: predictions that go beyond the established empirical data that a theory is introduced to explain, such as Einstein’s prediction (derived from the theory of relativity) that light rays traveling to Earth from distant stars would be deflected by the gravitational force of the sun. These predictions are especially important in adjudicating conflicts between competing theories that explain the same range of empirical data. Thus, the conflict between the corpuscularian (particle) and wave theories of light, which both explained and predicted the established laws of reflection, refraction, and the rectilinear propagation of light, was adjudicated by Foucault’s experiment, which demonstrated that light decelerates when moving from a less dense to a denser medium (for example, from air to water), as the wave theory predicted, but contrary to the acceleration predicted by the corpuscularian theory. Such adjudicating instances are often characterized as **crucial instances** or **crucial experiments**.

Both the inductivist and hypothetico-deductivist positions have proved popular with scientific psychologists. It would be hard to find a more forceful advocacy of inductive ascent than Watson’s characterization of behaviorist psychology in *Behaviorism* (1924) or a clearer illustration of hypothetico-deductivism than the system of theoretical postulates and derived empirical predictions in Hull’s *Principles of Behavior* (1943).

PHILOSOPHY AND PHYSIOLOGY

Many of the early theorists discussed in this work, such as Aristotle, Augustine, Aquinas, Descartes, Hume, and Kant, are usually characterized as philosophers rather than psychologists, and their work forms the basis of courses in philosophy and history of philosophy. However, although such early theorists were deeply concerned with **ontological** and **epistemological questions** about the fundamental nature of reality and our knowledge of it, they also developed substantive theories of perception, cognition, and behavior. Thus the Scottish philosopher David Hume (1711–1776), who argued that we cannot have knowledge of fundamental entities such as material substance or the self, developed detailed psychological explanations of how we come to believe in material bodies and enduring selves and conceived of his project as “an attempt to introduce the experimental method of reasoning into moral subjects” (Hume, 1739/1973).

To describe such theorists as philosophers or psychologists is anachronistic, since our conception of philosophy as a conceptual discipline and psychology as an empirical discipline is a product of the separate institutional development of the academic disciplines of philosophy and psychology in the early 20th century

(Reed, 1987). University academics concerned with both philosophical and psychological questions were designated as professors of philosophy for generations before, and Wundt held chairs in philosophy throughout his career at the universities of Heidelberg, Zürich, and Leipzig.

For this reason I have avoided trying to organize the historical narrative of this work in terms of traditional philosophical categories, such as the distinction between rationalists and empiricists or idealists and materialists. Theorists such as Descartes and John Locke (1632–1704) did disagree about whether we can have knowledge through pure reason. Descartes, the **rationalist**, maintained that we can have **a priori knowledge**, independently of experience, based upon reason; Locke, the **empiricist**, maintained that we can have only **a posteriori knowledge**, based upon experience. However, they also shared fundamental psychological theories about the imagistic nature of ideas and the conscious accessibility of all mental states.

Many of the early theorists discussed in this work are characterized as scientists or physiologists. Such characterizations are also anachronistic. Our conception of science as a social and professional institution, whose practitioners are committed to the empirical evaluation of formalized theories, is a relatively recent invention. The term *scientia*, from which the English term *science* is derived, originally referred to any form of knowledge, theoretical or practical. The science of physiology in its modern sense is tied to 18th- and 19th-century conceptions of cellular organization and the central and peripheral nervous system. The ancient Greeks originally used the term *physiology* to reference the study of nature in general; the medical restriction of the term to theories of nature employed to explain the functions of the human body was for centuries tied to the theory of the “four humors” (Hatfield, 1992).

However, we can recognize anticipations of our conception of science in the early naturalistic and mathematical speculations of the ancient Greeks, in the increasing emphasis on the empirical evaluation of theories from the 16th century onward, and in the development of scientific societies in the 17th century. Analogously, we can recognize discussions of the functions of the human organism and its components from the ancient Greeks to present-day physiologists. Although it is anachronistic to talk about early science and scientists and early physiology and physiologists, it is justified to the degree that many early thinkers developed theories about the structures and processes that still form part of the subject matter of contemporary sciences such as physiology.

The same is true of psychology. Although the institutional science of psychology is a late-19th-century creation (in Europe and America), many of the mental states and processes studied by contemporary psychologists (such as sensation, perception, emotion, memory, dreaming, learning, language, and thought) were objects of theoretical interest and empirical study for many early theorists or, as they are sometimes called in this work, proto-psychologists.

DISCUSSION QUESTIONS

1. Do you think that psychologists have more reason to study the history of their discipline than physicists or biologists? Is the history of psychology intrinsically more interesting because it is the history of attempts to attain a scientific understanding of our own mentality and behavior?
2. Could there be an internal contextualist history of psychology? Could there be an external presentist history of psychology? A history of psychology in terms of the prevailing zeitgeist is usually an external history, and a history of psychology in terms of the contributions of “great men” is usually internal. But must this be the case?
3. If a discipline is objective and advances causal explanations that are subject to systematic empirical evaluation, is that sufficient to constitute it as a scientific discipline? If not, what other features do you think are necessary?
4. Can you think of a psychological state or process that is not atomistic, not invariant in cultural space and historical time, and figures in causal explanations that are not universal?
5. Why do you think psychologists are so committed to experimentation? Should they be? Why do you think that social psychologists are so committed? Is experimentation especially suited to social psychology? Or especially problematic in social psychology?

GLOSSARY

a posteriori knowledge Knowledge based upon (after) experience.

a priori knowledge Knowledge independent of (prior to) experience.

atomism The principle that entities can be individuated and exist independently of other entities to which they may be related.

conceptual history A history of significant conceptual continuities and discontinuities in the development of a discipline.

contextualist history A history that attempts to explicate historical episodes and epochs neutrally in their own terms.

crucial instance/crucial experiment An empirical outcome enabling the adjudication of competing theories via their different predictions about the same empirical domain.

determinism The principle that there is a set of prior conditions sufficient for the production of any event, such that no other outcome is possible.

dogmatic empiricism The view that scientific theory and causal explanation are restricted to the description of the correlation of observables.

eliminative induction The method of eliminating competing causal hypotheses until only one viable hypothesis remains.

empiricist Someone who maintains that all knowledge is a posteriori, based upon experience.

epistemological question A question concerning our knowledge of reality. From the Greek *episteme*, meaning “knowledge.”

explanatory reduction The principle that the best explanation of a complex entity, property, or process is given by an analysis of its material components.

external history A history of a discipline in terms of (external) social, economic, political, and cultural conditions.

falsifiability The characteristic of a scientific theory that allows it to be falsified by observation; also called testability. According to Popper, the falsifiability of scientific theories is what distinguishes science from pseudoscience.

great men history A history that ascribes major developments to the influence of individuals.

historiography The theory and methodology of history.

hypothetico-deductive method The method of postulating theories and testing them via their deductive or predictive implications.

inductive method The method of deriving theories from observations.

instrumentalism The view that scientific theories “about” unobservable entities are not potentially true descriptions of them, but merely linguistic devices that facilitate the prediction of the behavior of observable entities.

internal history A history of a discipline in terms of the development of theories and methods within the discipline.

materialism The view that ultimate reality is material.

meaning empiricism The doctrine that linguistic items derive their meaning by association with—or their definition in terms of—observable entities.

method of agreement The methodological principle that if instances of an effect have only one condition in common, then that condition is the cause of the effect.

method of concomitant variation The methodological principle that if one condition increases or decreases while an effect increases or decreases, that condition is the cause of the effect.

method of difference The methodological principle that if an instance in which an effect occurs and an instance in which it does not occur differ with respect to only one condition, then that condition is the cause, or an essential part of the cause, of the effect.

methodological empiricism The principle that requires that all scientific descriptions and explanations be subject to observational evaluation.

novel predictions Predictions that go beyond the established empirical data that theories are introduced to explain and whose confirmation plays a significant role in establishing the theories.

objectivity 1. A characteristic of propositions when their truth or falsity is determined by independent facts. 2. A characteristic of the theoretical judgments of scientists when they are unbiased, when they are based only upon empirical evaluation.

ontological invariance The principle that kinds of entities in a scientific domain can be re-identified in all regions of space and time.

ontological question A question concerning the fundamental nature of reality. From the Greek *ontos*, meaning “being.”

operational definition A definition of the meaning of theoretical propositions in terms of observables.

presentist history A history in which a discipline is represented as approaching and approximating (idealized) contemporary theory and practice.

pseudoscience A discipline in which theoretical propositions are untestable or unfalsifiable.

psychological empiricism The doctrine that all concepts or ideas are derived from experience.

rationalist Someone who maintains that it is possible to have a priori knowledge, independently of experience, based upon reason.

realism The view that scientific theories about entities that are not observable are potentially true descriptions of them.

relational An entity is relational if it can be individuated and exist only in relation to other entities.

subjectivity A characteristic of the theoretical judgment of scientists when they are biased by individual or collective preferences or by social, political, or religious interests in the advocacy of certain theories.

universality of causal explanation The principle that one and the same causal explanation applies to every instance of a class of events, regularities, or structures.

zeitgeist history A history that ascribes major developments to the “spirit of the times.”

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Ancient Greek Science and Psychology

THE ORIGINS OF PSYCHOLOGICAL KNOWLEDGE MAY BE SAID TO BE AS old as humankind. From as early as recorded time, men and women have speculated about the nature and source of psychological states and processes and their relationship to human behavior. Theoretical reflections on sensation, memory, and dreaming, for example, are to be found in many ancient works, such as the Hindu sacred texts known as the Vedas (which precede the first millennium BCE) and the Assyrian “dream books” (from around the fifth millennium BCE).

Many early cultures, such as the Egyptian and Babylonian, tried to understand human psychology and behavior in terms of the activity of some immaterial “spirit” or “soul,” usually intimately associated with breath and with the action of the heart and lungs. The Greek term *psyche*, from which the term *psychology* is derived, is etymologically tied to words signifying breath (*pneuma*) or wind (Onians, 1958). There is nothing especially remarkable about this. At a basic level of observation, it is obvious that whatever enables the human organism to act in a purposive fashion is intimately associated with the action of the heart and lungs. When activity in these organs ceases, so also does the activity of the human organism.

Many early theories that postulated immaterial spirits or souls also maintained that such entities could enjoy a life after death in some spiritual realm. However, not all early theories were committed to the notion of an afterlife, and for those that were, it was often an impoverished and literally shady sort of thing. In Greek mythology, for example, the dead survived as shadows of their former selves, which could only be temporarily revived via blood sacrifice.

Beliefs in immaterial spirits or souls are often characterized as **animistic** and are to be found in many so-called primitive cultures today. However, we should guard against the rather condescending assumption that all earlier cultures explained mind and behavior in terms of immaterial spirits or souls and that humans came to a proper understanding of these matters only with the development of psychological science, since such assumptions can seriously prejudice our approach to the history of psychology. Although many ancient thinkers did